Summary of the Base Paper and of the Network Estimators

Diebold and Yılmaz (2014) define connectedness through the H-step generalized forecast-error variance decomposition (GFEVD) of an approximating VAR. The share of the forecast-error variance of variable i due to shocks in variable j is read as a weighted, directed link $j \to i$; row sums of the off-diagonal entries give how much each node receives ("from"), column sums how much it transmits ("to"), and the cross-sectional average is total connectedness. Using the generalized decomposition of Koop et al. (1996) and Pesaran and Shin (1998) makes the network invariant to the ordering of the variables. Rolling-window estimates track how connectedness changes over time.

NETS (Barigozzi and Brownlees, 2019) estimates the network directly as a sparse object: a lasso-type estimator of a VAR delivers a *Granger* network (lagged dependence), the sparse concentration matrix of the VAR innovations delivers a *contemporaneous* partial-correlation network, and their combination delivers a *long-run* partial-correlation network.

FNETS (Barigozzi et al., 2024) first removes a (possibly dynamic) common factor structure, estimated by dynamic principal components in a generalized dynamic factor model (Forni et al., 2000), and then estimates a sparse VAR and sparse (long-run) partial correlations for the idiosyncratic component. It therefore separates connectedness due to pervasive common shocks from idiosyncratic, network-type interdependence.

[To be completed: professor's papers on the idiosyncratic component, to be added]

1 Motivation

[To be completed: Why measuring connectedness among emerging equity markets matters (contagion, portfolio diversification, policy spillovers). Argentina is treated as one case among the 17 markets (its final role, decision 6 of paper structure v2, is still to be defined by the group). Suggested anchors from the data: the Argentine primary-election shock of August 2019 (largest weekly ARGT drop, -28.4% log return) and the March 2020 crash.]

2 Introduction

[To be completed: Research question (working version): how does measured connectedness among emerging equity markets change with the estimation method (Diebold–Yilmaz, sparse VAR, NETS, FNETS), and what share of it is a global common factor versus idiosyncratic? State hypotheses and the contribution; summarize the main results once Section 5 is written.]

3 Data

Universe and sample. We use US-listed, USD-denominated country exchange-traded funds (ETFs) as tradable proxies for 17 emerging equity markets: Argentina (ARGT), Chile (ECH), Brazil (EWZ), Mexico (EWW) and Peru (EPU) in Latin America; South Korea (EWY), Taiwan (EWT), India (EPI), China (FXI), Thailand (THD), Indonesia (EIDO), Malaysia (EWM), Vietnam (VNM) and the Philippines (EPHE) in Asia; South Africa (EZA), Turkey (TUR) and Poland (EPOL) in EMEA. The S&P 500 ETF (SPY) is included as a global reference node. Funds were selected if their price history starts by 3 March 2011 (7-day tolerance), at most 5% of weeks are missing, and average daily traded volume is at least USD 3 million. Colombian funds (ICOL, GXG), INDA and MCHI did not pass the criteria; India and China are represented by EPI and FXI. The common sample runs from 11 March 2011 to 23 October 2026, i.e. 816 weeks, with no missing values.

Returns and frequency. Prices are dividend- and split-adjusted, so returns are total returns. We use weekly Friday-to-Friday log returns. The weekly frequency follows the course indication that weekly returns retain some forecastability that is absent at daily frequency; it also mitigates non-synchronous trading between Asian and EMEA local markets and their US-listed ETFs. Unlike Diebold and Yılmaz (2014), who study (realized) volatilities, we study returns: no daily data are available to build realized weekly volatility, and returns are the object of the forecastability argument. Returns are not devoid of cross-market predictive content: in a full-sample VAR(1) on the 18 series, 59 of the 324 slope coefficients (18.2%) are significant at the 5% level, against 5% expected by chance. Our networks therefore describe the transmission of return shocks, not of risk.

Descriptive statistics. Table 1 reports summary statistics. All series are strongly leptokurtic and mostly negatively skewed; an augmented Dickey–Fuller test rejects a unit root for every series. Five weekly log returns exceed 25% in absolute value (ARGT in the week of the August 2019 primary elections and four observations in March 2020); they are genuine market moves and are kept. The March 2020 crash dominates the tails: of the ten largest weekly drops pooled across the 17 emerging markets, eight occur in the weeks ending 13 and 20 March 2020, and 14 of the 17 markets record their worst week in March 2020.

Table 1: Descriptive statistics of weekly log returns, 2011-03-11 to 2026-10-23 (816 weeks). Kurtosis is not in excess.

Ticker	Country	Region	Start	Mean (%)	Std (%)	Min (%)	Max (%)	Skew.	Kurt.
ARGT	Argentina	Latin America	2011-03-11	0.15	4.18	-28.4	24.0	-0.51	8.9
ECH	Chile	Latin America	2011-03-11	-0.03	3.56	-21.2	16.2	-0.36	5.6
EWZ	Brazil	Latin America	2011-03-11	0.01	4.26	-26.5	22.2	-0.18	6.4
EWW	Mexico	Latin America	2011-03-11	0.07	3.29	-24.0	10.3	-0.74	7.8
EPU	Peru	Latin America	2011-03-11	0.12	3.35	-19.3	13.3	-0.39	5.9
EWY	South Korea	Asia	2011-03-11	0.14	3.66	-20.3	16.0	-0.40	6.6
EWT	Taiwan	Asia	2011-03-11	0.23	2.85	-12.6	11.2	-0.43	4.4
EPI	India	Asia	2011-03-11	0.09	2.90	-18.8	13.3	-0.35	6.3
FXI	China	Asia	2011-03-11	0.01	3.47	-12.6	17.0	0.03	4.5
THD	Thailand	Asia	2011-03-11	0.07	2.91	-23.5	9.6	-0.85	9.0
EIDO	Indonesia	Asia	2011-03-11	-0.08	3.68	-33.6	20.7	-1.25	14.6
EWM	Malaysia	Asia	2011-03-11	0.01	2.25	-12.2	10.9	-0.22	6.5
VNM	Vietnam	Asia	2011-03-11	-0.01	3.15	-19.3	11.9	-0.35	5.9
EPHE	Philippines	Asia	2011-03-11	0.02	2.92	-19.3	14.4	-0.57	7.4
EZA	South Africa	EMEA	2011-03-11	0.07	4.06	-26.9	14.1	-0.57	6.5
TUR	Turkey	EMEA	2011-03-11	-0.03	4.60	-23.0	20.0	-0.63	6.1
EPOL	Poland	EMEA	2011-03-11	0.08	3.66	-25.1	15.2	-0.58	6.5
SPY	United States	Global	2011-03-11	0.25	2.23	-15.7	11.4	-0.74	9.3

Limitations. (i) ETF returns embed exchange-rate movements against the USD and are subject to non-synchronous trading. (ii) ARGT changed its underlying index in August 2014 (from a Solactive Argentina index to the MSCI All Argentina 25/50). (iii) FXI tracks only the 50 largest Chinese firms (FTSE China 50) and EPI is the earnings-weighted WisdomTree India fund, so neither is a broad market index. (iv) Colombia is absent. (v) Liquidity requirements and the common start date exclude younger funds. (vi) Without daily data, realized-volatility connectedness cannot be computed.

4 Methods

Let $y_t \in \mathbb{R}^N$ be the vector of weekly returns ($N = 18$). All methods are estimated on rolling windows of $w = 104$ weeks; each estimate is dated at the last week of its window. For the comparison, all methods use the same 179 windows (step of four weeks, first window ending 1 March 2013).